DATA ANALYTICS PORTFOLIO

Ecommerce Sales Performance Analysis

A four-year review of revenue, profitability, customer behavior, and operational patterns

TOTAL REVENUE	TOTAL PROFIT	ORDERS ANALYZED
$11.37M	$2.22M	30,000

Analysis period: January 2023 - December 2026 | Source: ecommerce_orders workbook

Executive summary

The dataset contains 30,000 ecommerce orders across four full calendar years. It generated $11.37 million in revenue and $2.22 million in profit, an overall profit margin of 19.5%. The analysis highlights where revenue is concentrated, what customers prefer, and where improvement opportunities exist.

- **Revenue peak.** 2023 was the strongest year, producing $2.96 million in revenue. August was the top month across the full period at $1.07 million.
- **Category leader.** Electronics delivered $6.32 million in revenue and $1.04 million in profit - more than half of total sales.
- **Geographic opportunity.** Saudi Arabia was the top country at $1.20 million, while Dubai was the leading city at $1.21 million.
- **Customer experience.** Average customer rating was 4.05/5 and average delivery time was 4.44 days. Standard shipping handled 16,420 orders.

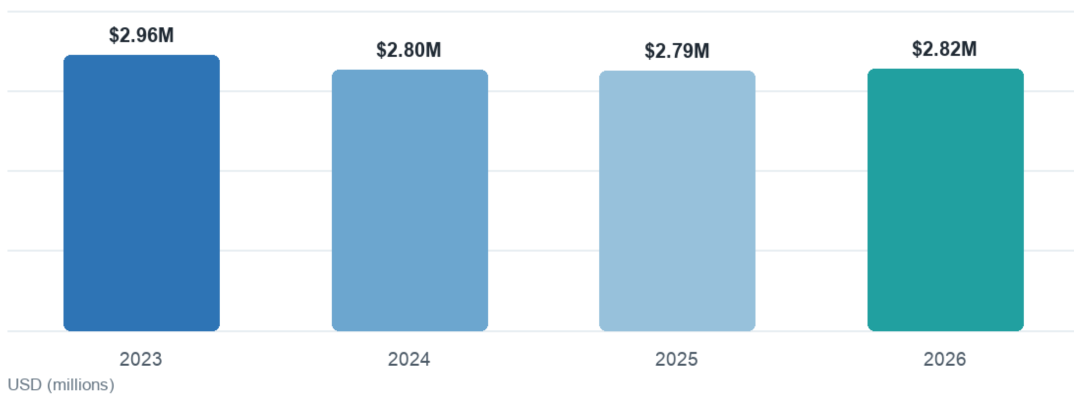
Business questions answered

Question	Answer
How much revenue did the business generate?	$11,370,043.99 across four years
Which year performed best?	2023, with $2,962,427.11 in revenue
Which product category leads?	Electronics: $6,320,059.31 revenue
Which market is strongest?	Saudi Arabia: $1,196,810.50 revenue
What is the return-risk category?	Fashion, with a 13.87% return rate

Revenue and profitability performance

Annual revenue was broadly stable, with 2023 establishing the high-water mark. The following three years varied within a narrow band, indicating a resilient sales base but also a need for growth initiatives to regain the 2023 peak.

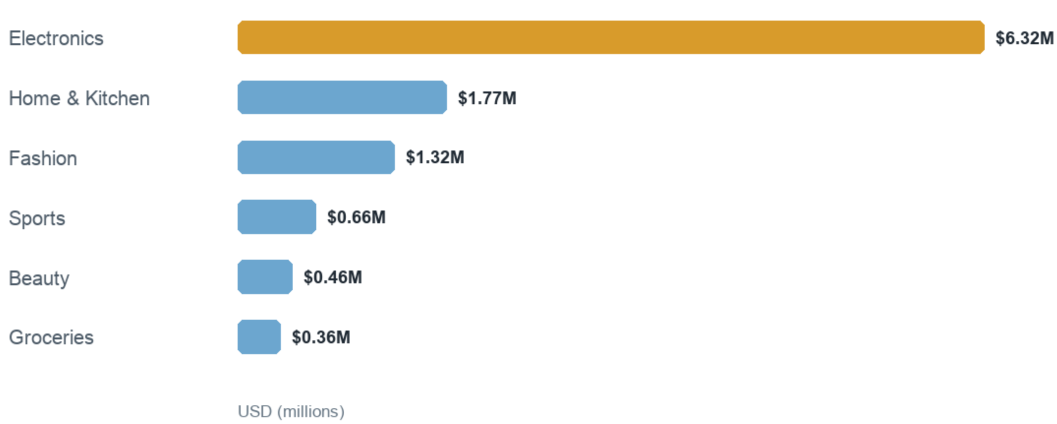
Annual revenue



Category contribution

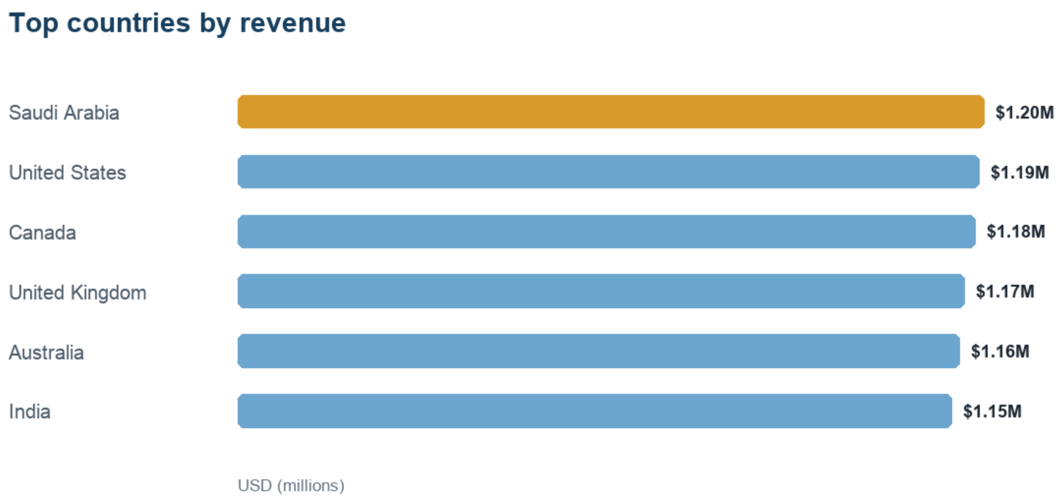
Electronics is the primary revenue engine, contributing 55.6% of total revenue. It also produces the most profit ($1,040,688.00), making it the category to protect and scale. Fashion leads unit volume (18,821 units), but its return rate requires attention.

Revenue by product category



Markets and customer behavior

Revenue is well distributed across countries, yet Saudi Arabia and the UAE city of Dubai lead the market view. This suggests an opportunity to examine the customer segments, traffic sources, and product mix that perform best in these markets.



Customer, payment, and fulfillment signals

Metric	Key finding
Most-used payment method	Credit Card - 5,080 orders
Highest-revenue payment method	PayPal - $1,938,263.16
Most-used shipping method	Standard - 16,420 orders
Largest traffic source	Social Media - $1,981,282.37 revenue
Most frequent customer	CUST000031 - 14 orders

Top customers by lifetime spending

Customer ID	Total spending
CUST000459	$10,712.12
CUST006906	$10,179.74
CUST000025	$9,308.43
CUST000381	$9,035.20
CUST005303	$8,789.97

Operational insights and recommendations

What the data indicates

- **Discount discipline.** Discount percentage and profit have a weak negative correlation (-0.153). Deep discounts are not the only profit driver, but tighter discount governance can protect margin.
- **Returns.** Fashion has the highest return rate at 13.87%, above the next-highest category (Books at 10.91%). Fit guidance, product imagery, and return-reason analysis should be prioritized.
- **Seasonality.** August is the highest-revenue month ($1,071,268.49); Wednesday is the top day by revenue ($1,663,976.90). Campaign timing can lean into these periods.
- **Delivery experience.** An average delivery time of 4.44 days alongside a 4.05/5 rating gives a practical baseline for tracking service-level improvements.

Portfolio-ready recommendations

Priority	Recommendation	Why it matters
1	Protect Electronics margin	It accounts for the largest share of revenue and profit.
2	Reduce Fashion returns	Fashion leads unit sales but has the highest return rate.
3	Scale winning markets	Study Saudi Arabia and Dubai to replicate successful mix and acquisition channels.
4	Use seasonal campaign windows	Prioritize August and high-performing midweek periods for promotional tests.

Method note

Metrics were calculated from the ecommerce_orders data table. Revenue is the sum of Order_Amount; profit is the sum of Profit_Amount. Return rate is the percentage of orders marked Returned = Yes. The source workbook contains 30,000 populated orders; one blank record was excluded from the analysis.